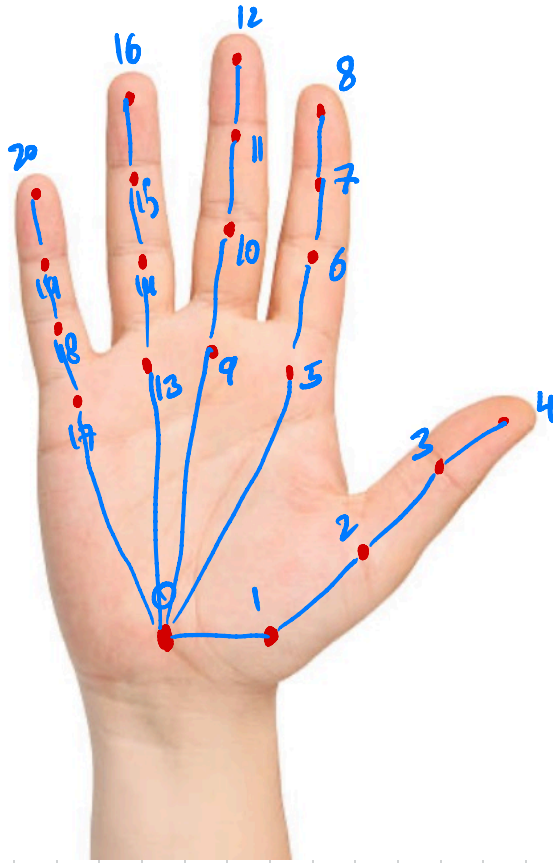




Overall landmarks from pipeline.



each landmark is a 3D point (x, y, z)

x is horizontal position $0-1 \Rightarrow$ left-right

y is vertical position

est. using neural network.

z is depth relative to wrist (how far landmark from camera)

z small $\Rightarrow$ close to cam, large far from cam

$z=0$ same depth as wrist

-0.2 very close to camera

-0.05 slightly close

$+0.2$ very far from camera

$+0.05$ slightly far.

normalization $\Rightarrow$ make wrist (point #0)

as our system origin $(0,0,0)$.

so for each landmark point P we do

$p = p - \text{wrist}$ (including wrist so now its $(0,0,0)$)

now we want to remove scale so hand size matters

less, we do that by dividing by scale which

is the dist between wrist and middle finger (12)

$\text{scale} = \text{landmark}[12].\text{length} \ \& \ p = p / \text{scale}.$

now small scale should be ignored cuz it indicates
issues in recognizing hand position (detection glitch)
or frame noise.